Securing Your Snowflake Account

Kyro Mills
Account Engineer

© 2026 Snowflake Inc. All Rights Reserved

Safe Harbor and Disclaimers

Other than statements of historical fact, all information contained in these materials and any accompanying oral commentary (collectively, the “Materials”), including statements regarding (i) Snowflake’s business strategy, plans or priorities, (ii) Snowflake’s new or enhanced products, services, and technology offerings, including those that are under development or not generally available, (iii) market growth, trends, and competitive considerations, (iv) our vision for Snowpark, the Data Cloud, and industry-specific Data Clouds, including the expected benefits and network effects of the Data Cloud; and (v) the integration, interoperability, and availability of Snowflake’s products, services, or technology offerings with or on third-party platforms or products, are forward-looking statements. These forward-looking statements are subject to a number of risks, uncertainties and assumptions, including those described under the heading “Risk Factors” and elsewhere in the Annual Reports on Form 10-K and the Quarterly Reports on Form 10-Q that Snowflake files with the Securities and Exchange Commission. In light of these risks, uncertainties, and assumptions, the future events and trends discussed in the Materials may not occur, and actual results could differ materially and adversely from those anticipated or implied in the forward-looking statements. As a result, you should not rely on any forwarding-looking statements as predictions of future events. Forward-looking statements speak only as of the date the statements are first made and are based on information available to us at the time those statements are made and/or management's good faith belief as of that time. Except as required by law, we undertake no obligation, and do not intend, to update the forward-looking statements in these Materials.

Any future product or roadmap information (collectively, the “Roadmap”) is intended to outline general product direction. The Roadmap is not a commitment, promise, or legal obligation for Snowflake to deliver any future products, features, or functionality; and is not intended to be, and shall not be deemed to be, incorporated into any contract. The actual timing of any product, feature, or functionality that is ultimately made available may be different from what is presented in the Roadmap. The Roadmap information should not be used when making a purchasing decision. In case of conflict between the information contained in the Materials and official Snowflake documentation, official Snowflake documentation should take precedence over these Materials. Further, note that Snowflake has made no determination as to whether separate fees will be charged for any future products, features, and/or functionality which may ultimately be made available. Snowflake may, in its own discretion, choose to charge separate fees for the delivery of any future products, features, and/or functionality which are ultimately made available.

The Materials may contain information provided by third-parties. Snowflake has not independently verified this information, and usage of this information does not mean or imply that Snowflake has adopted this information as its own or independently verified its accuracy.

© 2026 Snowflake Inc. All rights reserved. Snowflake, the Snowflake logo, and all other Snowflake product, feature and service names mentioned in the Materials are registered trademarks or trademarks of Snowflake Inc. in the United States and other countries. All other brand names or logos mentioned or used in the Materials are for identification purposes only and may be the trademarks of their respective holder(s). Snowflake may not be associated with, or be sponsored or endorsed by, any such holder(s).



Security Overview

© 2026 Snowflake Inc. All Rights Reserved

Snowflake Layered Security

(Above the Cloud Provider)



1.

Network Access

Restrict access to specific IP addresses or range with Network Policies

Optionally: Limit data traffic to Cloud Providers Private Network (VPC)

2.

Authentication (AuthN)

To verify identity of users/processes using username/password, Single Sign-On (SSO), Key Pair, or OAuth.

3.

Authorization (AuthZ)

Rights to access objects in Snowflake using roles and privileges.

4.

Continuous Data Protection

Protect data with AES-256 bit encryption, yearly online re-keying of data, Time Travel (90 days), undrop, Fail-safe, data masking, row-access policies, and external data tokenization.

© 2026 Snowflake Inc. All Rights Reserved

What We'll Build Today

Five controls, in the order we build them today — each one closes a specific gap.

Layer	What you'll build
Setup	Two demo users: a human and a service account that still uses a password
Network	A network rule and policy that limit the service account to trusted IPs, plus Snowflake-managed SaaS rules
Authentication	Two policy tiers: MFA for humans, key pair only for service accounts
Trust Center	Run a scanner and read a real finding: a service user that still uses a password
Remediate	Fix the finding, re-scan, and watch it clear

© 2026 Snowflake Inc. All Rights Reserved

Meet Cortex Code

Snowflake's AI coding agent — right where you already work.

What it is

An agentic assistant for Snowflake: write and run SQL, build pipelines, debug, and explore your data in plain language — in the CLI, in Snowsight, and on the desktop.

Why it matters

It turns the RBAC you're about to learn into instant productivity: ask in plain language, and it writes SQL against the roles and objects you're allowed to see.

You enable it today

Access is a role grant — the same RBAC pattern from this session. We grant it in the notebook, then you self-enable in 2 steps.



Account Recovery

© 2026 Snowflake Inc. All Rights Reserved

Break-Glass Emergency Access

What is a Break-Glass Account?

A dedicated emergency admin account — like the fire-alarm glass you only break in a true lockout — created before any restrictive policy so you always have a way back in.

01. Early Setup

Created first — stood up before SSO, MFA, network, or auth policies, so nothing can lock it out as you add layers.

02. Minimal Dependencies

Password + OTP only, restricted to Snowsight UI (CLIENT_TYPES = (SNOWFLAKE_UI)) — no SSO/IdP dependency.

03. Resilient Policy

Its own user-level network policy — user-level overrides account-level, so it survives an account-wide network lockout.

04. Secure Vaulting

ACCOUNTADMIN granted, credentials stored in a vault; OTP codes generated once and captured immediately.

Why it matters: If SSO/MFA misfires or an account policy locks everyone out, this is your guaranteed recovery path — no Snowflake Support ticket required.

© 2026 Snowflake Inc. All Rights Reserved

Source: Break Glass Admin



Network Management

Network Policies

© 2026 Snowflake Inc. All Rights Reserved

Network Rules

- Network rules are schema-level objects that group network identifiers into **logical units**.
- The following features use network rules to control network traffic:
  - Network policies use network rules to control inbound network traffic to the Snowflake service and internal stages.
  - External network access uses network rules to restrict access to external network locations from a Snowflake UDF or procedure.

Syntax

```
CREATE [ OR REPLACE ] NETWORK RULE <name>
TYPE = { IPV4 | AWSVPCEID | AZURELINKID |
HOST_PORT | PRIVATE_HOST_PORT }
VALUE_LIST = ( '<value>' [, '<value>', ... ] )
MODE = { INGRESS | INTERNAL_STAGE | EGRESS }
[ COMMENT = '<string_literal>' ]
```

Example

```
CREATE NETWORK RULE allow_list
TYPE = IPV4
MODE = INGRESS
VALUE_LIST = ('47.88.25.32/27');
```

** With IPV4 USES CIDR NOTATION

Source: Network Rules

© 2026 Snowflake Inc. All Rights Reserved

Network Policy

- Network policies **limit connections to your Snowflake account** to a predefined list of allowed IP addresses or ranges.
- They help mitigate the risk of leaked credentials being used from untrusted locations.
- Can be set at an **account** or **user** level

Syntax

```
CREATE [ OR REPLACE ] NETWORK POLICY [ IF NOT EXISTS ] <name>
  [ ALLOWED_NETWORK_RULE_LIST = ('<network_rule>' ,...,
'<network_rule>' )]
  [ BLOCKED_NETWORK_RULE_LIST = ('<network_rule>' ,...,
'<network_rule>' )]
  [ COMMENT = '<string_literal>' ]
```

Example

```
CREATE NETWORK POLICY access policy
  ALLOWED_NETWORK_RULE_LIST =
('allow list')
  BLOCKED_NETWORK_RULE_LIST =
('deny_list');

ALTER ACCOUNT SET NETWORK_POLICY =
access_policy;
ALTER USER <my_user> SET
NETWORK_POLICY = access_policy;
```

Source: Network Policy

© 2026 Snowflake Inc. All Rights Reserved



Authentication

Authentication Policies

© 2026 Snowflake Inc. All Rights Reserved

Authentication Policy

- Authentication policies provide you with **control over how a client or user authenticates** by allowing you to specify:
  a. Whether users must enroll in multi-factor authentication (MFA).
  b. Which authentication methods require multi-factor authentication.
  c. The allowed authentication methods, such as SAML, passwords, OAuth, key pair authentication, and programmatic access tokens.

Syntax

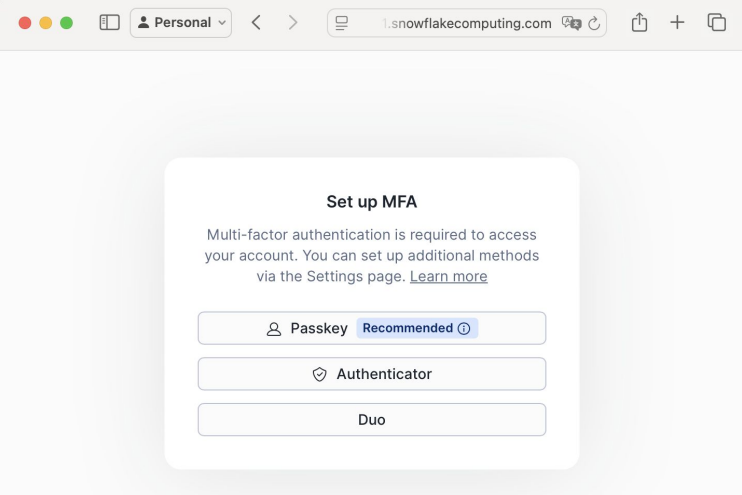
```
CREATE OR ALTER AUTHENTICATION POLICY <name>
  [ AUTHENTICATION_METHODS = ( '<string_literal>' , ... ) ]
  [ MFA_ENROLLMENT = { 'REQUIRED' | 'REQUIRED_PASSWORD_ONLY' } ]
  [ MFA_POLICY = ( ALLOWED_METHODS = ( { 'ALL' | 'PASSKEY' | 'TOTP' | 'OTP' |
'DUO' } )
      [ , ENFORCE_MFA_ON_EXTERNAL_AUTHENTICATION = { 'ALL' | 'NONE' } ] ) ]
  [ CLIENT_TYPES = ( '<string_literal>' , ... ) ]
  [ SECURITY_INTEGRATIONS = ( '<string_literal>' , ... ) ]
  [ PAT_POLICY = ( {list_of_properties} ) ]
  [ WORKLOAD_IDENTITY_POLICY = ( {list_of_properties} ) ]
  [ COMMENT = '<string_literal>' ]
```

© 2026 Snowflake Inc. All Rights Reserved

Source: Authentication Policy

MFA

- When a user is enrolled in MFA, they are required to use an MFA method as a **second factor of authentication**. Snowflake allows the following MFA methods:
  a. Authenticating with a passkey that can be stored and accessed in a variety of ways.
  b. Authenticating with an authenticator app that generates a time-based one-time passcode (TOTP).
  c. Authenticating with Duo.



Example

```
CREATE OR ALTER AUTHENTICATION POLICY
require_mfa_policy
  AUTHENTICATION_METHODS = ('PASSWORD')
  MFA_ENROLLMENT = 'REQUIRED'
  MFA_POLICY = (ALLOWED_METHODS = ('PASSKEY'));
```

**MFA is intended for *users* who authenticate with a password

Source: Snowflake - MFA

Source: Snowflake - MFA Setup

© 2026 Snowflake Inc. All Rights Reserved

SSO (Single Sign-On)

In a federated environment, authentication is separated from access; an external **Identity Provider (IdP)** verifies who the user is, then passes that trust to **Snowflake (SP)** so users sign in once.

Two Roles

Service Provider (SP): Snowflake

Identity Provider (IdP): Creates, maintains, and authenticates credentials.

Native Support

Direct support for **Okta** and **Microsoft Entra ID**.

Works with any SAML 2.0-compliant IdP (Google, Ping, etc.) via custom app.

Login Flows

SP-initiated: Start at Snowflake.

IdP-initiated: Start at IdP portal.

SAML2_ENABLE_SP_INITIATED = TRUE

Configuration

Requires a SAML2 security integration:

```
CREATE SECURITY INTEGRATION ...
TYPE = SAML2
```

Logout & Timeouts

IdP-dependent: Standard vs. Global logout (Entra ID supports Global).

IdP timeout does not end active Snowflake sessions.

Best Practices

Pair with an **authentication policy** allowing SAML.

Do not set local passwords for SSO-only users.

© 2026 Snowflake Inc. All Rights Reserved



Authorization

Access Control

Source: Snowflake Access Control Overview

© 2026 Snowflake Inc. All Rights Reserved

Snowflake Access Control Methods

As compared to other database systems, Snowflake Roles are the constructs used to govern access, not Users and Roles. So, you can't grant object rights to users other than thru roles. Users just pick a Role to use to access Snowflake objects.

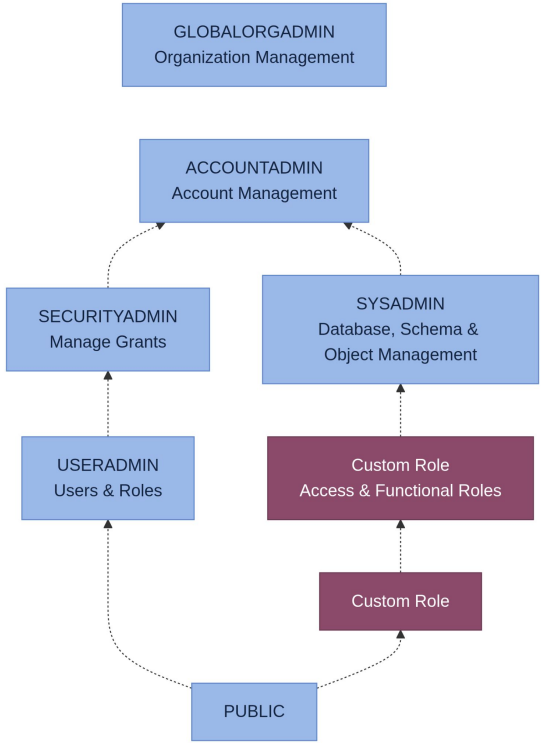
Role-Based Access Control (RBAC)
Access privileges are assigned to Roles, which are in turn assigned to Users

Discretionary Access Control (DAC)
Each object has an owner, who can in turn grant access to that object

FUTURE grant privilege
Enables grantees to have access to future created objects.

© 2026 Snowflake Inc. All Rights Reserved

System Roles



Key point: Role privileges are inherited from roles above that role in the hierarchy.

GLOBALORGADMIN
Manages the organization from the organization account — account lifecycle (create, rename, drop), organization-level billing and usage, and organization-wide listings/data sharing. Granted to a few trusted users.

ACCOUNTADMIN
A powerful role for ACCOUNT level configuration and management that inherits all other role capabilities. Should not be used day-to-day and limit to 2-4 users.

SECURITYADMIN
A role with MANAGE GRANTS privilege that can alter **any** other role or grant in the Account. It inherits user management from USERADMIN role.

USERADMIN
For day-to-day management of User and Role creation and removal. Limited to roles and users it is the owner of.

SYSADMIN
Used primarily to create schema objects and Grant access to Roles. Can use roles that it has been granted and the data controlled by those roles. (Data Masking being an exception.)

PUBLIC
Pseudo-role automatically granted to every user and role. Note: Everyone has privilege to use Public schema because they will have PUBLIC role rights.

© 2026 Snowflake Inc. All Rights Reserved

System Roles — What Each One Does

Always use the lowest role that does the job — privileges inherit upward.

Role	What it does	When to use it
GLOBALORGADMIN	Manages the org from the org account — account lifecycle, org billing, org-wide sharing	Org-level tasks only; a few trusted users
ACCOUNTADMIN	Top of the account; inherits all other roles	Setup + account config only; limit to 2–4 users
SECURITYADMIN	MANAGE GRANTS — can alter any role or grant in the account	Account-wide grant / role changes
USERADMIN	Creates and manages the users and roles it owns	Day-to-day user and role admin
SYSADMIN	Creates objects; custom roles roll up here	Building databases, schemas, warehouses
PUBLIC	Pseudo-role automatically granted to every user and role	Baseline access everyone shares

ACCOUNTADMIN Role Guidelines



The ACCOUNTADMIN Role is powerful - restrict its use

- Intended for initial set up and ongoing administration, not day-to-day operations
- Ongoing Administration:
  - Configuring Account level parameters
  - Monitoring of usage and billing and configuring Resource Monitors
  - Managing incoming and outgoing Shares
  - Manage sessions / running queries

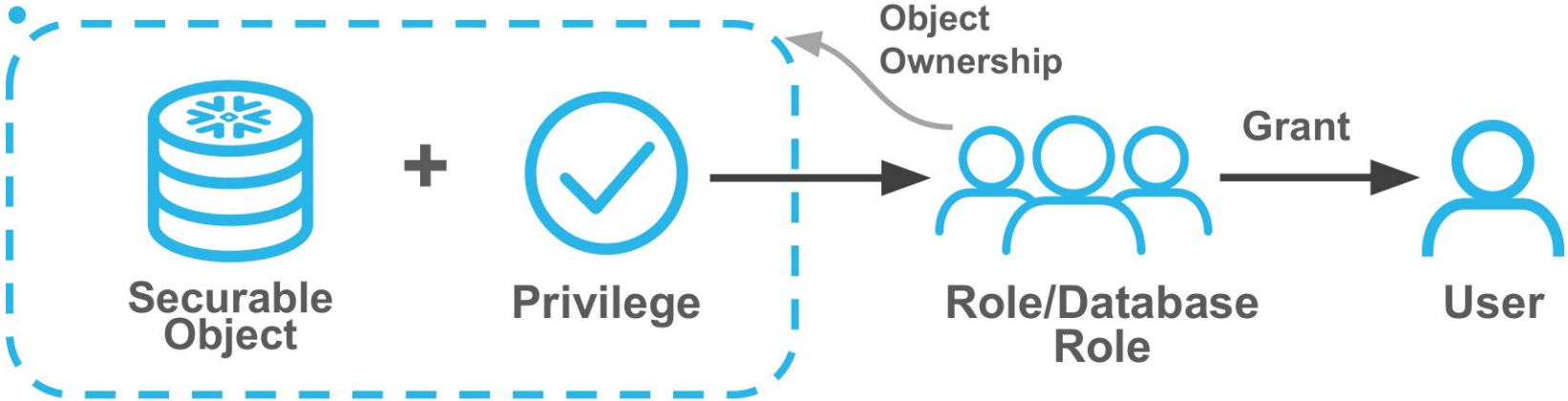
Recommended practices

- Assign ACCOUNTADMIN to limited users (minimum 2 for no single point of failure)
- Require MFA on all users with this role (allow and have username/password as backup to SSO)
- If access is lost, Snowflake security protocols could make a reset take up to 2 working days
- Avoid ACCOUNTADMIN as default role for any user
- Do not create objects with the role ACCOUNTADMIN

© 2026 Snowflake Inc. All Rights Reserved

Access Control Concepts

- A **securable object** is an entity to which access can be granted. Unless allowed by a grant, access is denied. e.g. Data (database, schema, table, task, stream, stored procedure, function) & Warehouse.
- A **Persona** denotes the different hats a human can wear in order to perform business functions.
- A **Role** is an entity to which privileges can be granted. Roles can be broadly grouped into Admin, Engineering, and Analytical.
- An **Access Role** is a role which is granted privileges to perform specific operations on a **securable object**. e.g. of such privileges include Execute, Read, Write, Manage, Usage etc.
- A **Functional Role** is a role granted to a user and has Access roles granted to it.
- A **User** in Snowflake is either a human or a Service Account.



© 2025 Snowflake Inc. All Rights Reserved

SECURITYADMIN

1. Create User
2. Create Role
3. Manage Grants

Can manage **all** users and grants in Account.
Use USERADMIN instead if you can.

USERADMIN

1. Create User
2. Create Role

Can **only** manage users/grants created by USERADMIN. Use for day-to-day admin.

© 2026 Snowflake Inc. All Rights Reserved

Custom Roles



01

Define Role

Define custom roles that reflect your organization's structure

02

Least Privileges

Assign minimum necessary permissions to each role.

03

Security & Compliance

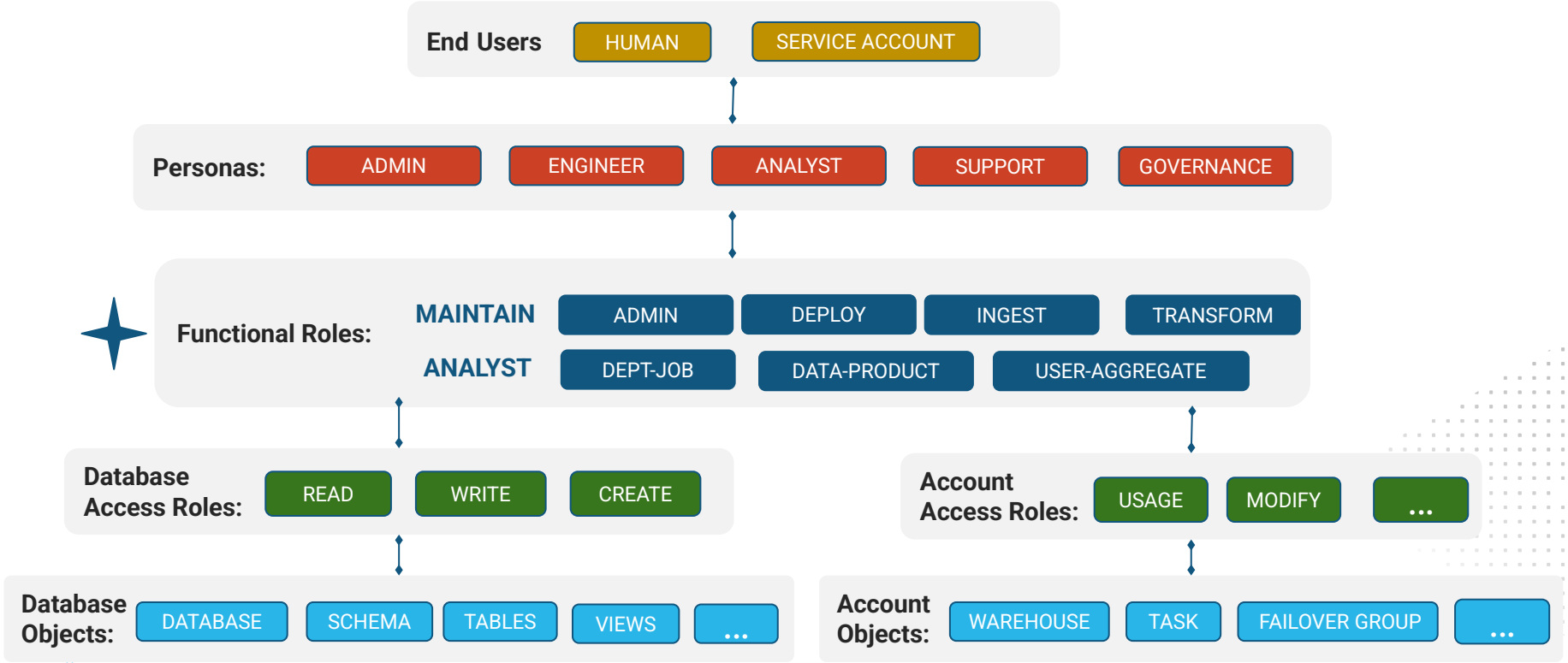
Regularly monitor/audit role assignments and permission changes

Source: Snowflake - RBAC

© 2026 Snowflake Inc. All Rights Reserved

BIG PICTURE

WHERE FUNCTIONAL ROLES COME IN



© 2026 Snowflake Inc. All Rights Reserved

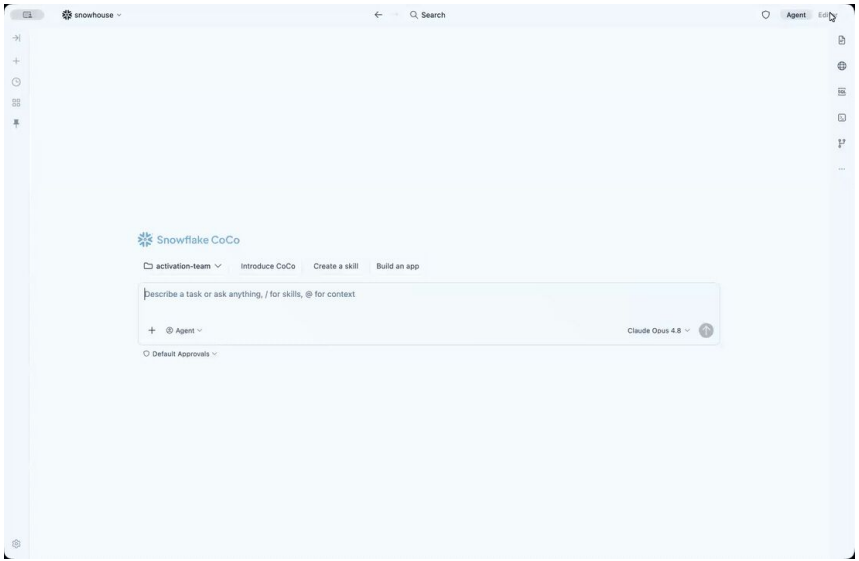
Turn On Cortex Code — In 2 Steps

Step 1 — Get access

An admin grants the Cortex Code roles to your role: COPILOT_USER + CORTEX_USER (or CORTEX_AGENT_USER). We run exactly this grant in today's notebook.

Step 2 — Download & sign in

Get the desktop app (or open it in Snowsight), sign in with your Snowflake login, and run your first prompt.



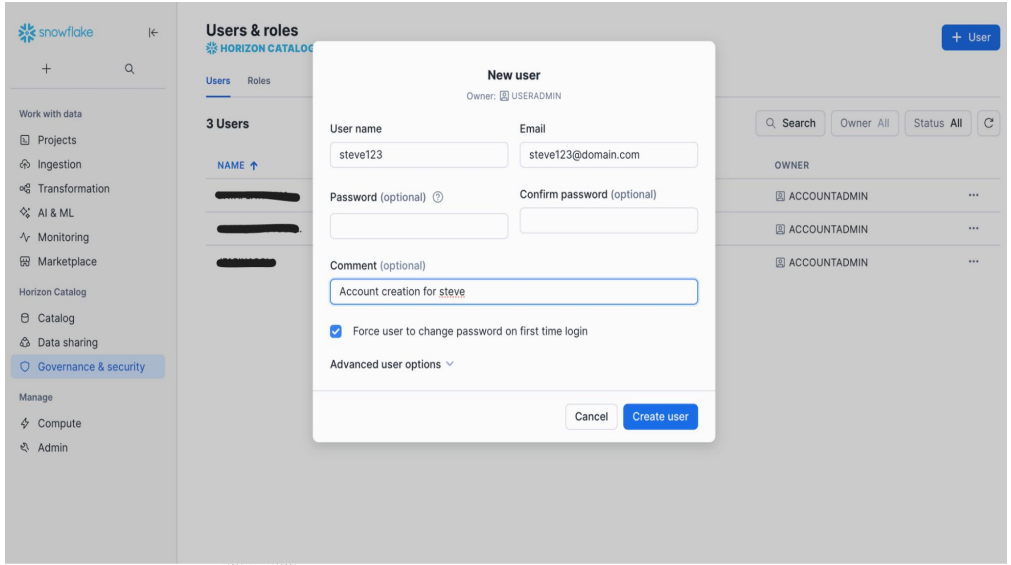


User Management

User Provisioning

© 2026 Snowflake Inc. All Rights Reserved

Manual User Provisioning



Syntax

```
CREATE [ OR REPLACE ] USER [ IF NOT EXISTS ] <name>
  [ objectProperties ]
  [ objectParams ]
  [ sessionParams ]
  [ [ WITH ] TAG ( <tag_name> = '<tag_value>' [ , <tag_name> = '<tag_value>' , ... ] )
```

Example

```
CREATE USER STEVE123
EMAIL = 'steve123@domain.com'
DEFAULT_ROLE = myrole
DEFAULT_SECONDARY_ROLES = ('ALL') ;
```

Key parameters to set

TYPE — PERSON vs SERVICE; drives which MFA / authentication policies apply.
LOGIN_NAME — the sign-in identifier, separate from the user's display name.
DEFAULT_ROLE — the role a user assumes automatically at login.

[Source: Snowflake Create User]

Automatic User Provisioning (SCIM)

Custom SCIM integrations allow users to build their own applications to interface with their identity provider to provision, map, and manage users and roles to Snowflake.

SCIM roles in Snowflake must own any users or roles that are imported from the identity provider. If the Snowflake SCIM role does not own the imported users or roles, updates in the identity provider will not be synced to Snowflake. Snowflake SCIM roles correlate with their identity provider (IdP):

- Okta SCIM Role: `okta_provisioner`
- Microsoft Entra ID SCIM Role: `aad_provisioner`
- Custom SCIM Role: `generic_scim_provisioner`

Source: Snowflake SCIM support

Source: Microsoft Entra ID SCIM

© 2026 Snowflake Inc. All Rights Reserved



Time for the Lab!

© 2026 Snowflake Inc. All Rights Reserved

Appendix — Security Best Practices Checklist

- Choose your user management strategy (e.g., SCIM)
- Choose your authentication strategy (e.g., SSO)
- Choose your network restriction strategy (e.g., corporate IPs / VPN)
- Configure SSO provider (if applicable)
- Configure SCIM provider (if applicable)
- Create users (directly in Snowflake or via SCIM)
- Set user type (PERSON, SERVICE)
- If using SSO, do not set a local password — but keep at least 1 ACCOUNTADMIN on a local password to troubleshoot SSO
- Ensure the TYPE attribute is set on all users
- Create authentication policies to enforce MFA on human users
- Create an admin authentication policy to avoid being locked out
- Create authentication policies for service users
- Create a session policy
- Create network policies for service users
- Create network policies at the account level
- Apply authentication and network policies to service users
- Apply password and session policies at the account level
- Apply the authentication policy at the account level
- Apply the network policy at the account level

Source: Security Whitepaper

THANK YOU

Before you go

Turn on Cortex Code — 2 steps, using the same grant you ran in today's notebook.

© 2026 Snowflake Inc. All Rights Reserved